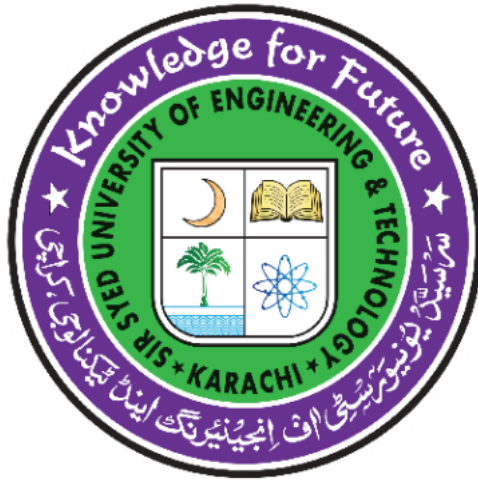


Project Report

WEB TECHNOLOGIES LAB (CS316L)

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Project: Chef Master – Full-Stack Recipe Management System

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Project Report: Chef Master

– Full-Stack Recipe Management System

Course: Web Engineering / Database Systems **Technology Stack:** MERN (MongoDB, Express.js, React.js, Node.js)

1. Introduction

ChefMaster is a dynamic, full-stack web application designed to revolutionize the way users discover, organize, and interact with cooking recipes. In an era where decision fatigue affects daily meal planning, ChefMaster provides an intelligent solution by allowing users to search for recipes based on ingredients they already possess, reducing food waste and decision time.

Initially developed as a static prototype, the project has evolved into a fully dynamic **Single Page Application (SPA)**. It now utilizes a **MongoDB database** to store recipes, user profiles, and feedback, ensuring data persistence and scalability. The application bridges the gap between raw ingredients and finished meals through a responsive, user-centric interface that adapts to the user's environment (e.g., Day vs. Late Night modes).

2. Scope

The scope of this project encompasses the design, development, and deployment of a modern web application using the **Client-Server Architecture**.

- **Functional Scope:** The system allows users to browse a vast database of recipes, create secure accounts, manage personalized favorite lists, scale ingredient quantities for different serving sizes, and submit feedback.
- **Data Scope:** Transitioning from static JSON files to a **NoSQL Database (MongoDB)** allows for real-time data management. The system handles structured data for Recipes (ingredients, steps, nutrition) and unstructured data for User Sessions and Feedback.
- **Technical Scope:** The project demonstrates mastery of:

- **Frontend:** React.js for component-based UI and State Management.
- **Backend:** Node.js and Express.js for RESTful API creation.
- **Database:** MongoDB for persistent storage and querying.
- **Security:** JSON Web Tokens (JWT) for authentication and Bcrypt for password hashing.

3. Detailed Feature Analysis

ChefMaster distinguishes itself through a blend of technical complexity and polished User Experience (UX). Below are the core features:

1. Database-Driven Recipe Architecture

Unlike static websites, ChefMaster fetches its content dynamically from a MongoDB database.

- **Dynamic Loading:** Recipes are no longer hardcoded. The application queries the backend API (/api/recipes) to retrieve the latest data.
- **Scalability:** This architecture allows the system to hold thousands of recipes without impacting frontend performance, as data is fetched only when needed.

2. Secure User Authentication & Persistence

The application implements a robust security layer to manage user identities.

- **JWT Authentication:** Users can securely Sign Up and Log In. Upon login, the server issues a **JSON Web Token**, allowing the user to remain authenticated across page refreshes.
- **Password Encryption:** User passwords are encrypted using **Bcrypt** before being stored in the database, ensuring high-level security.
- **Persistent Profiles:** A personalized dashboard greets the user by name (e.g., "Hello, Sana").

3. Syncing Favorites to Cloud

A major feature upgrade is the "Cloud Sync" capability.

- **Logic:** When a logged-in user marks a recipe as a "Favorite" (❤️), this action is sent to the API and saved in the user's document in MongoDB.

- **Benefit:** This ensures that a user's favorite recipes are never lost, even if they close the browser or switch devices.

4. Context-Aware "Late Night Mode"

ChefMaster features an intelligent "Dark Mode" that goes beyond simple color changes.

- **Visual Adaptation:** A toggle switch activates a high-contrast dark theme with neon accents, designed to reduce eye strain in low-light environments.
- **Smart Filtering Algorithm:** When active, the backend logic automatically filters the recipe results. It hides complex, time-consuming dishes and exclusively displays "**Comfort Foods**" (Burgers, Pizzas) and "**Quick 5-Minute**" recipes, catering specifically to late-night cravings.

5. Interactive "Cooking Mode" with Portion Scaling

To assist users during the actual cooking process, recipes open in a focused Modal View.

- **Distraction-Free UI:** The background is blurred to help the user focus on instructions.
- **Dynamic Math Engine:** Users can select serving sizes (1x, 2x, 4x). The application uses a custom algorithm to parse ingredient strings (e.g., "2 eggs") and mathematically scale them in real-time (e.g., "8 eggs") without breaking the text format.

6. User Feedback System

A dedicated communication channel connects the user to the database.

- **Form Handling:** Users can submit suggestions or bug reports via a responsive form.
- **Data Storage:** These submissions are captured via a POST request and stored in a dedicated feedbacks collection in MongoDB for future review.

4. Conclusion

The development of ChefMaster has successfully met the objectives of creating a full-stack, database-backed web application. By migrating from local file storage to **MongoDB**, the project achieved true data persistence and dynamic content management.

This project highlights the power of the **MERN Stack** in building modern web solutions. It seamlessly integrates complex backend logic (Authentication, API routing, Database queries) with a responsive and aesthetic frontend. ChefMaster stands as a functional, scalable prototype ready for future enhancements such as AI-based meal recommendations or mobile app integration.